

Keywords:

Injection molding, experimental data, viscosity, pressure, machine data, Euromap77, Acrylonitrile Butadiene Styrene (ABS)

Overview

This dataset contains the data of 68 injection molding experiments containing information about machine setting parameters, time-resolved viscometer pressure data as well as data recorded by several internal sensors of an injection machine. The dataset contains 17 experiments with unique machine settings. Each experiment includes 4 repeated cycles (runs), to take into account statistical deviations.

The data was collected by SKZ (German Plastics Center, Würzburg) during the research project "LoKI", funded by the "Bayerisches Verbundforschungsprogramm (BayVFP) des Freistaates Bayern" (Project ID: DIK-2209-0027// DIK0468/01, Project Management Agency VDI).

For data acquisition the experiments were conducted under following conditions:

An injection molding machine (Model: 160-750PX, Manufacturer: KraussMaffei Technologies GmbH) was equipped with a viscometer (Model: MDT465FXL-1/2-2M-A, Manufacturer: Dynisco Europe GmbH) at the injection nozzle. In a series of experiments, standard parts (see Figure 1) were molded from ABS material (Name: Terluran GP-35, Manufacturer: INEOS Styrolution Europe GmbH) with different machine settings. For each experiment, timeseries pressure data from the viscometer were recorded during each cycle. Additionally, after each cycle, the corresponding EUROMAP77 machine data was acquired via the OPC UA interface of the machine. Furthermore, machine setting parameters were also included in the data for each cycle.

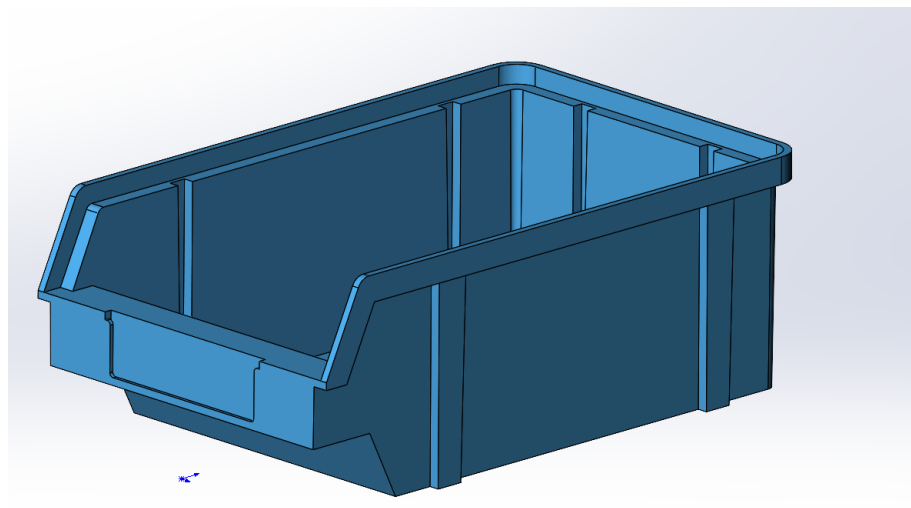


Figure 1 Geometry of standard parts used in the injection experiments. Part dimensions are approximately 100 mm x 160 mm x 60 mm

Motivation

The data was collected to test the feasibility of training machine learning models to predict material viscosity parameters during injection, based on internal OPC UA machine data, with the goal to build an AI based viscometer.

Furthermore, a second motivation was the training of machine learning models, which can predict modified machine settings, to compensate for material viscosity fluctuations between cycles, e. g. when recycle thermoplastic materials are used for injection molding, which often show batch-wise variations of material properties.

Data Structure

The data is structured into three sub datasets, `set_parameter_and_EM77_data`, `viscometer_pressure_data` and `quality_table_data`, provided as both, .csv and .parquet files.

Each data column in the dataset is named according to the following general conventions:

Metadata (META_*) – Metainformation, e.g. experiment identifier named `META_experiment`.

Set Parameters (SET_*) – Configured machine settings for each experiment, e. g. cylinder temperature setting named `SET_CylinderTemperature`.

Measured Data (MEAS_*) – Recorded machine and pressure data for each experiment, e.g. measured cushion volume named `MEAS_EM77_CushionVolume`

Data of all subsets can be merged by using the metadata identifiers **META_experiment** and **META_run**, which are included in each subset. These two identifiers are, when used in combination, unique identifiers for each injection cycle contained in the datasets.

Sub dataset 1: `set_parameter_and_EM77_data`

This subset contains all scalar data, such as machine setting parameters and recorded Euromap77 machine data for each injection cycle. Euromap77 data are structured according to the [Euromap77 Standard](#), which describes the interface between injection molding machines (IMM) and manufacturing execution systems (MES) to enable standardized and efficient data exchange. This ensures interoperability between machines and software systems across different manufacturers

It contains the following columns:

Metadata (META_)

META_experiment: Unique experiment identifier.

META_run: Number of the cycle within each experiment.

META_JobCycleCounter_global: Global job cycle counter according to machine cycle counter.

META_CycleStart: Start cycle number of experiment, according to machine cycle counter.

META_CycleEnd: End cycle number of experiment, according to machine cycle counter.

SET PARAMETERS (SET_)

SET_CylinderTemperature: Cylinder temperature setting. Unit is [°C]

SET_HotRunner/Nozzle: Hot runner/nozzle temperature. Unit is [°C]

SET_InjectionSpeed: Configured injection speed. Unit is [ccm/s]

SET_UP: Injection switchpoint. Kept constant during experiments. Unit is [ccm]
Kept in Data for completeness

SET_MoldTemperature: Mold temperature setting. Unit is [°C]

Measured EUROMAP77 Data (MEAS_EM77_*)

MEAS_EM77_BarrelTemperatureZone 1-6: Temperatures of different barrel zones. 6 columns, one for each temperature zone. Unit is [°C]

MEAS_EM77_CushionStroke: Measured cushion stroke. Unit is [mm]

MEAS_EM77_CushionVolume: Measured cushion volume. Unit is [cm³]

MEAS_EM77_CycleTime: Total cycle time recorded. Unit is [s]

MEAS_EM77_DosingTime: Time taken for dosing. Unit is [s]

MEAS_EM77_FlowIndex: Measured flow index. Unit is [cm³/s]

MEAS_EM77_HoldHydraulicPressureMaximum: Maximum hold hydraulic pressure. Unit is [bar]

MEAS_EM77_HoldSpecificPressureMaximum: Maximum hold specific pressure. Unit is [bar]

MEAS_EM77_HydraulicPressureMaximum: Maximum hydraulic pressure. Unit is [bar]

MEAS_EM77_InjectionSpeedAverage: Average injection speed recorded. Unit is [mm/s]

MEAS_EM77_InjectionSpeedMaximum: Maximum injection speed. Unit is [mm/s]

MEAS_EM77_InjectionTime: Injection time recorded. Unit is [s]

MEAS_EM77_PlastificationHydraulicPressureMaximum: Maximum plastification hydraulic pressure. Unit is [bar]

MEAS_EM77_PlastificationRotationalSpeedMaximum: Maximum plastification rotational speed. Unit is [rpm]

MEAS_EM77_PlastificationSpecificPressureMaximum: Maximum plastification specific pressure. Unit is [bar]

MEAS_EM77_SpecificPressureMaximum: Maximum specific pressure. Unit is [bar]

MEAS_EM77_TransferSpecificPressure: Specific pressure at transfer. Unit is [bar]

MEAS_EM77_TransferStroke: Stroke at transfer. Unit is [°C]

MEAS_EM77_TransferVolume: Volume at transfer. Unit is [°C]

Sub data set 1 characteristics:

Size: 68 rows × 34 columns.

Data Types:

8 columns: Integer (int64).

26 columns: Floating point (float64).

Completeness: No missing values in the dataset.

Sub dataset 2: viscometer_pressure_data

This subset contains the timeseries viscometer pressure data for each cycle. Each row represents a distinct time point, which can be linked to its corresponding experiment and run using the metadata columns. Within the same experiment and run, successive time points are recorded in chronological order, beginning at 0 seconds.

Metadata (META_)

META_experiment: Unique experiment identifier.

META_run: Number of the cycle within each experiment.

Measured Pressure Data (MEAS_)

MEAS_time_s: Cycle time. Unit is [s]

MEAS_pressure_fronsensor_bar: Time dependent pressure measured during injection cycle at front of viscometer tube (nozzle side). Unit is [bar].

MEAS_pressure_backsensor_bar: Time dependent pressure measured during injection cycle at back of viscometer tube (mold side). Unit is [bar].

MEAS_pressure_difference_bar: Difference of pressure_fronsensor and pressure_backsensor. Unit is [bar]

Sub data set 2 characteristics:

Size: 1048575 rows × 6 columns.

Data Types:

2 columns: Integer (int64).

4 columns: Floating point (float64).

Completeness: No missing values in the dataset.

Sub dataset 3: quality_table_data

This subset contains data from the Quality Table of the injection machine. The Quality Table contains several parameters, measured by internal sensors of the injection machine for each cycle, such as pressure and energy parameters. The Quality Table data is used by injection molding experts to detect problems and find error causes of the process.

Metadata (META_)

META_experiment: Unique experiment identifier.

META_run: Number of the cycle within each experiment.

META_CycleStart: Start cycle number of experiment, according to machine cycle counter.

META_CycleEnd: End cycle number of experiment, according to machine cycle counter.

Quality Table Data (MEAS_KM_*)

MEAS_KM_Zyklus- Zeit Z [s]: Total cycle time. Unit is [s]

MEAS_KM_Plast- Zeit Z [s]: Plasticizing time. Unit is [s]

MEAS_KM_Spritz- Zeit Z [s]: Injection time. Unit is [s]

MEAS_KM_Umsch MasDrZ [bar] : Switching pressure. Unit is [bar]

MEAS_KM_Max WkzDr1 [bar] : Maximum tool cavity pressure. Unit is [bar].

MEAS_KM_Max F WkzDr1 [bar] : Maximum force applied by the mold pressure. Unit is [bar].

MEAS_KM_Fläche WkzDr1 [bar*s] : Integrated tool cavity pressure over time (pressure-time integral). Units are [bar*s]

MEAS_KM_Max F Massedr [bar] : Maximum mass pressure during plasticizing. Unit is [bar].

MEAS_KM_Fläche Massedr [bar*s] : Integrated mass pressure over time. Unit is [bar].

MEAS_KM_Max StauDr [bar] : Maximum backpressure during plasticizing. Unit is [bar].

MEAS_KM_Max F Spritzge [mm/s] : Maximum injection speed in mm/s. A key factor in filling the mold efficiently. Units are [mm/s]

MEAS_KM_Fläche Spritzge [mm/s*s] : Integral of the injection speed over time, representing the total speed exposure. Units are [mm/s²]

MEAS_KM_Max MasDr [bar] : Maximum machine pressure (Maschinendruck) during injection. Unit is [bar].

MEAS_KM_Energie ZylHtg [Wh] : Energy consumption of the cylinder heating elements in watt-hours. Units are [Wh].

MEAS_KM_Energie HptAntr [Wh] : Energy consumption of the main drive motor. Units are [Wh].

MEAS_KM_Energie PeriAuto [Wh] : Energy used by auxiliary peripherals and automation systems. Units are [Wh].

MEAS_KM_Energie Gesamt [Wh] : Total energy consumption of the injection molding cycle. Units are [Wh].

MEAS_KM_Max F Leistung [kW] : Maximum power demand during the process. Units are [kW].

MEAS_KM_Fläche Leistung [Wh] : Energy consumed over time. Units are [Wh].

MEAS_KM_Fläche Spritzweg [mm*s] : Integral of the injection stroke over time. Units are [mm*s].

MEAS_KM_Masse- Polster [cm³] : Residual melt cushion volume. Unit is [cm³].

Sub data set 3 characteristics:

Size: 68 rows × 25 columns.

Data Types:

9 columns: Integer (int64).

16 columns: Floating point (float64).

Completeness: No missing values in the dataset.